DICTO Kyushindo Cores

Towers and spires round exact cores in Fibonacci beats: the DICTO Cleo Towers, the DICTO Polar Cleo Towers, the OG core of 80 Aktis, the OTT core of 38 rays (Oktaktis, Triaktis and Tetraktis), and the OG turned out

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Version 1 · 21 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/551887e/studies (research branch, commit 551887e: folders 2026-10-20-cleo-towers, 2026-10-20-og-spires, 2026-10-20-ott-rays and 2026-10-21-og-turned-out, with the exact tools of 2026-10-19-five-fold-octet and 2026-10-18-ocd-octet; also attached to this record)
The ideas (shells of towers in a Fibonacci rhythm round the CLEO sphere; their polar; the OG and OTT cut into rays from the centre; turning the OG's rays out with the golden step) and the names Cleo Towers, Polar Cleo Towers, Hex Aktis, Deltoid Aktis, OG, OTT, Oktaktis, Triaktis, Tetraktis, the OG turned out and the Kyushindo series are DICTO's; computation, exact proofs and drafting with Claude (Anthropic). The rule of each step is that of Universal PiTet [1], the three-dimensional antipedal construction; its reverse is Smarandache's face pedal polyhedron [2, 9].



Abstract. DICTO's Kyushindo series builds outward from an exact core, shell by shell, with towers. One *beat* is one step of the Universal PiTet rule [1], T(S) = {x : v·x ≤ v·v for every corner v of S}; shells fuse 1, 1, 2, 3 and 5 beats, and each shell is cut into towers, one over each face of the solid inside it (the cone from the centre over that face, cut off by the solid outside). We prove in exact arithmetic in ℚ(√5): the **DICTO Cleo Towers**, round the CLEO sphere (the icosidodecahedron), are five shells of 32, 30, 32, 32 and 60 towers, each kind of tower one shape and each shell filled exactly. Reciprocation about the sphere of radius² 15 − 5√5 takes beat k to beat 5 − k, and turns them into the **DICTO Polar Cleo Towers**: shells of 5, 3, 2, 1 and 1 beats from the centre, with 30, 32, 30, 60 and 32 towers. The DICTO Ogdoecontahedron (OG) cut from its centre gives the **OG core of 80 Aktis**: 20 **DICTO Hex Aktis** and 60 **DICTO Deltoid Aktis**, each set one shape, taking (65 − 3√5)/110 and (45 + 3√5)/110 of the OG. The DICTO OTT gives a core of 38 OTT rays: 8 **DICTO Oktaktis**, each exactly a Cleo-Icos and lying in a hexagon plane of the OG, 24 **DICTO Triaktis** and 6 **DICTO Tetraktis**. DICTO's **OG turned out** flips the 80 Aktis outward at 1/φ of their height, tips at φ times the foot points of the OG's faces: then beat k is exactly φ^k times the k-th solid of the OG's downward family [2], every solid holds the one before, and the shells of 1, 1 and 2 beats hold 80, 132 and 140 towers of 2, 3 and 3 shapes, each shell filled exactly. The boundary solids change their surface pattern from shell to shell; on the TQH the trapezoids pair into 30 ribbons on the two-fold axes, which DICTO sees as a possible new shell structure. The shells of 3 and 5 beats of the turned-out sequence are still being computed. All claims are exact except where marked.

1. Beats, shells and towers

The beat. As in Universal PiTet [1]: for a convex solid S centred at the origin, T(S) = {x : v·x ≤ v·v for every corner v of S}. Each face of T(S) passes through a corner of S at right angles to the line from the centre. A corner v of S stays on the surface of T(S) exactly when w·v ≤ w·w for every corner w of S

(otherwise the face of w cuts it off). This is not true of every convex solid, but it holds, exactly, for every station stepped from in this paper: the ID, RT, GTI and FPD, and so for all their scaled copies, since the condition does not change with scale (it holds too for the OG, TPH, TQH and TQHD, but not for the TQP, where 60 of the 140 corners would be cut off). In the plane this is the classical antipedal construction about a point; T is its three-dimensional form [1, 10], and the reverse step P, whose corners are the foot points of S's face planes (the points of the planes nearest the centre), is Smarandache's face pedal polyhedron [9], run from the OG in [2]. DICTO calls one step a beat.

The rhythm. DICTO's idea: shells of towers as tall as possible, in a Fibonacci rhythm. Shell n fuses F_n beats, 1, 1, 2, 3, 5, so the shells end at beats 0 | 1 | 2 | 4 | 7 | 12, and the beats inside a shell are passed through, not drawn.

The towers. A shell between an inner solid A and an outer solid B ⊃ A is cut into one tower per face of A: the cone from the centre over that face, cut off by B, without the pyramid from the centre to the face (these are the shell pieces of [1]). A tower is one shape per symmetry orbit of faces; we prove for every shell below that the towers on each kind of face form one orbit of the 60 exact rotations of the icosahedral group (so they are congruent), have one exact volume, and add up to the volume of the shell with remainder exactly 0; we also check that the inner and the outer solid of every shell are each fixed by all 60 rotations. Corners are proved as in [1]: floating point only guesses which faces meet at a corner; every corner then solves its faces exactly and lies inside every face, every face holds at least three corners, and every edge lies in exactly two faces with $V - E + F = 2$.

2. The DICTO Cleo Towers (exact)

The core is DICTO's CLEO sphere [7, §5.5]: the icosidodecahedron (ID), cut from its centre into 20 CLEOs [6] on its triangles and 12 pentagonal pockets. Here it has circumradius φ (edge 1), the frame of [1]. Its beats are the stations of Universal PiTet: S_0 = ID, S_1 = RT (rhombic triacontahedron), S_2 = DICTO GTI (Golden Truncated Icosahedron), S_3 = DICTO FPD (Flat Pentakis Dodecahedron), and S_4 = c·ID with $c = 10/\varphi^4 = 35 - 15\sqrt{5} \approx 1.459$: one turn of four beats returns the ID exactly c times larger (checked again here), so $S_{k+4} = c \cdot S_k$ for every k [1].

Shell	Beats	From → to	Towers	Each tower (exact; ≈)	Shell volume
1	1	ID → RT	20 on triangles 12 on pentagons	$(45 - 17\sqrt{5})/120 \approx 0.058224$ $5/12 \approx 0.416667$	$(75 - 17\sqrt{5})/6 \approx 6.1645$
2	1	RT → GTI	30 on rhombi	$(3 - \sqrt{5})/4 \approx 0.190983$	$(45 - 15\sqrt{5})/2 \approx 5.7295$
3	2	GTI → c·ID	20 on hexagons 12 on pentagons	$(-236973 + 105982\sqrt{5})/12 \approx 0.829699$ $(1164825 - 520925\sqrt{5})/24 \approx 0.053701$	$(1124745 - 502955\sqrt{5})/6 \approx 17.2384$
4	3	c·ID → c·FPD	20 on triangles 12 on pentagons	≈ 0.998686 ≈ 3.613281	≈ 63.3331
5	5	c·FPD → c³·ID	60 on triangles	≈ 5.135315	≈ 308.1189

Table 1. The Cleo Towers, at ID circumradius φ. Shell 3 passes through the FPD, shell 4 through c·RT and c·GTI, shell 5 through c²·ID, c²·RT, c²·GTI and c²·FPD. The exact values of shells 4 and 5 (denominators 261, 9 and 27) are in tower-shells-exact.json.

So the five shells hold 32, 30, 32, 32 and 60 towers, 186 in all, and the solid of 12 beats is the ID again, $c^3 \approx 3.11$ times as wide as the core. Shells 1 and 2 are the first two shells of [1]; shells 3, 4 and 5, which fuse beats, are new. In each fused shell the towers stand on the faces of the shell's inner solid only, so the solids passed through leave no trace: shell 3, for instance, has towers on the GTI's 20 hexagons and 12 pentagons that reach to the corners of c·ID.

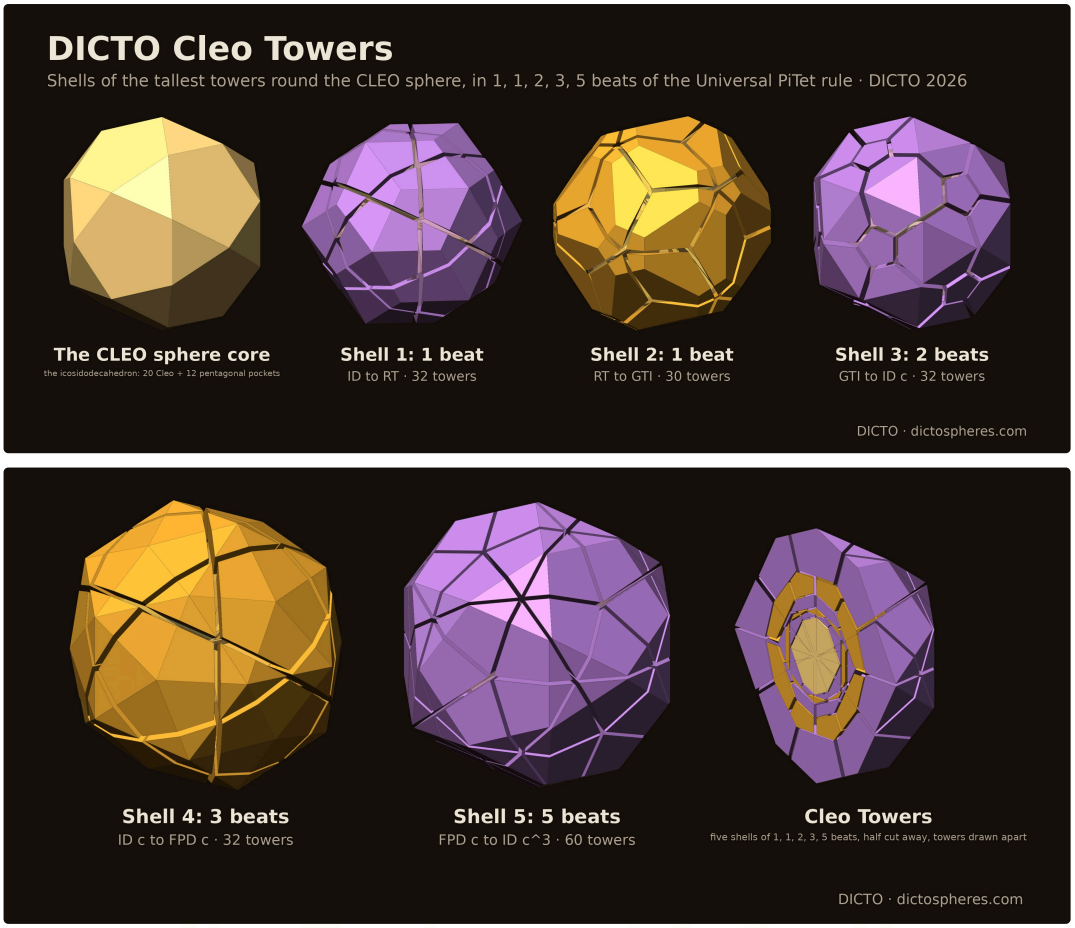


Figure 1. The DICTO Cleo Towers. The core (cream) and each shell's towers on their inner solid, drawn slightly apart, the shells alternately purple and gold; last, all five shells together with half cut away, the towers drawn apart. The labels c and c^3 are 10/φ^4 and its cube.

3. The DICTO Polar Cleo Towers (exact)

DICTO asked what the polar of the Fibonacci shells would give. Take the sphere through the GTI's corners, radius² $\rho^2 = 15 - 5\sqrt{5} = c\cdot\varphi^2$: $10/\varphi^4$ times the CLEO sphere's radius². Polar reciprocation about it takes every beat to another beat: $\mathbf{polar(S_k) = S_{5-k}}$, checked exactly for k = 0, …, 5 (ID ↔ c·RT, RT ↔ c·ID, GTI ↔ FPD; this is the self-polarity of [1, Proposition (iv)]); since polar(c·S) = polar(S)/c, it holds for every k. The boundaries 0 | 1 | 2 | 4 | 7 | 12 of the Cleo Towers go to 5 | 4 | 3 | 1 | −2 | −7: read from the centre, shells of **5, 3, 2, 1 and 1 beats**, the most beats deepest. The core is $S_{-7} = RT/c^2$, which cuts from its centre into 30 rhombic rays.

Shell	Beats	From → to	Towers	Each tower (exact; ≈)	Shell volume
1	5	RT/c² → GTI/c	30 on rhombi	(155284 + 69441√5)/1500000 ≈ 0.207039	(155284 + 69441√5)/50000 ≈ 6.2112
2	3	GTI/c → RT	12 on pentagons 20 on hexagons	(51911 − 17827√5)/83520 ≈ 0.144260 (146143 + 12334√5)/348000 ≈ 0.499203	(6343 − 741√5)/400 ≈ 11.7152
3	2	RT → FPD	30 on rhombi	(12982 − 5800√5)/27 ≈ 0.474286	(129820 − 58000√5)/9 ≈ 14.2286
4	1	FPD → c·ID	60 on triangles	(77875 − 34825√5)/27 ≈ 0.145655	(1557500 − 696500√5)/9 ≈ 8.7393
5	1	c·ID → c·RT	20 on triangles 12 on pentagons	(334125 − 149425√5)/3 ≈ 0.180821 (201250 − 90000√5)/3 ≈ 1.294008	(9097500 − 4068500√5)/3 ≈ 19.1445

Table 2. The Polar Cleo Towers, in the same frame; every kind of tower one shape, every shell filled exactly.

The polar shells hold 30, 32, 30, 60 and 32 towers, 184 in all. Shell 4 is the last shell of [1]. Shell 5 is the polar image of the Cleo Towers' first shell and, because polarity swaps the ID and the RT, it is that same shell c times larger: its towers are those of Table 1, shell 1, scaled by c (volumes by c^3).



Figure 2. The DICTO Polar Cleo Towers. The core, the rhombic triacontahedron RT/c² cut into 30 rhombic rays (cream), and each shell's towers on their inner solid, alternately purple and gold; last, all five shells with half cut away. "RT x c^-2" is RT/c², and so on.

4. The OG core of 80 Aktis (exact)

DICTO's idea: cut the DICTO Ogdoecontahedron (OG) [3], with its 20 regular hexagons and 60 kites, into its 80 face pyramids, every tip at the centre, as the CLEO sphere is cut into CLEOs and pockets. DICTO named them **Aktis** (Greek, ray): 20 **DICTO Hex Aktis** on the hexagons and 60 **DICTO Deltoid Aktis** on the kites. Values are in the frame of [4], where the OG has 102 corners and volume $(-1000 + 1240\sqrt{5})/1539 \approx 1.151868$.

- The 80 Aktis fill the OG exactly (volumes add up with remainder 0; neighbouring Aktis meet only in faces, as pyramids over different faces of a convex solid from an inside point must; the first 60 of the 180 neighbouring pairs were also intersected exactly).
- The 20 Hex Aktis are one shape, and the 60 Deltoid Aktis are one shape (corner matching: tip to tip, base corners in order, all distances equal).
- **Hex Aktis:** volume $(-2 + 2\sqrt{5})/81 \approx 0.030520$, height $\sqrt{((6 + 2\sqrt{5})/27)} \approx 0.622782$; the 20 take $(65 - 3\sqrt{5})/110 \approx 52.99\%$ of the OG.
- **Deltoid Aktis:** volume $(-4 + 8\sqrt{5})/1539 \approx 0.009024$, height $\sqrt{((42 + 2\sqrt{5})/109)} \approx 0.652955$; the 60 take $(45 + 3\sqrt{5})/110 \approx 47.01\%$.

The kites lie farther out than the hexagons, so the Deltoid Aktis are the taller rays.

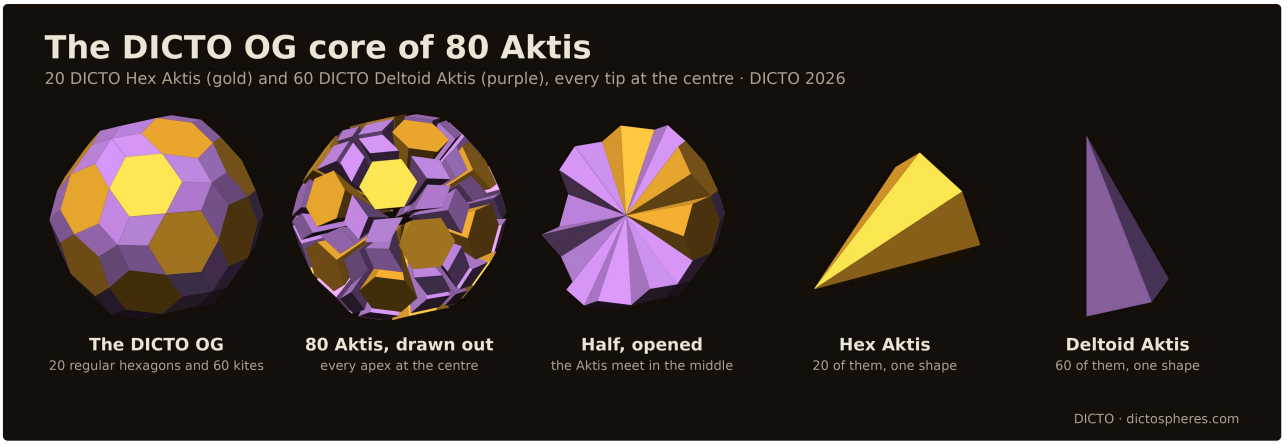


Figure 3. The OG core: the DICTO OG; its 80 Aktis drawn out, every tip at the centre; half of them, opened; one Hex Aktis (gold) and one Deltoid Aktis (purple).

5. The OTT core of 38 rays (exact)

The DICTO OTT (Octahedrally Truncated Triacontahedron) [5] is the rhombic triacontahedron cut by its octahedron at all 8 key sites: 24 corners, 38 faces (8 equilateral triangles in the octahedron's faces, 24 half-rhombi and 6 whole rhombi), volume $(-608 + 320\sqrt{5})/81 \approx 1.327676$, pyritohedral symmetry. DICTO's idea: cut it, like the OG, into its face pyramids from the centre, the **OTT rays**, which DICTO named **Oktaktis** (on the octahedron cuts), **Triaktis** (on the half-rhombi) and **Tetraktis** (four-sided, on the whole rhombi). The 38 rays fill the OTT exactly and fall into three shapes (corner matching):

Rays	Base	Each ray (exact; ≈)	Height ≈	Share of the OTT
8 Oktaktis	equilateral triangle (octahedron cut)	$(-4 + 4\sqrt{5})/81 \approx 0.061040$	0.6228	36.78%
24 Triaktis	half-rhombus	$(-16 + 8\sqrt{5})/81 \approx 0.023315$	0.6667	42.15%
6 Tetraktis	whole rhombus	$(-32 + 16\sqrt{5})/81 \approx 0.046631$	0.6667	21.07%

Table 3. The OTT core, in the frame of [4, 5]. The Triaktis and Tetraktis have one height because all 30 of their bases lie in the triacontahedron's face planes.

The Oktaktis are Cleo-Icos. DICTO noticed that the Oktaktis looks like a Cleo-Icos; we prove it exactly: each Oktaktis is exactly similar to the Cleo-Icos [7, §5.6], the pyramid from an icosahedron's centre to one face: $\text{height}^2/\text{edge}^2 = \varphi^4/12$ and $\text{side}^2/\text{edge}^2 = (\varphi^2 + 1)/4$, as for the icosahedron. **They lie in the OG's hexagon planes.** In the frame of [4], where both solids are given, each of the OTT's 8 octahedral triangles lies exactly in one of the 20 hexagon planes of the OG; so an Oktaktis has exactly the height of a Hex Aktis, $\sqrt{((6 + 2\sqrt{5})/27)}$. Since the OG lies inside the OTT [5], each of those 8 hexagons of the OG lies in an octahedral triangle of the OTT; more exactly, the hexagon's circumradius equals the triangle's inradius (squared, $(6 - 2\sqrt{5})/27$), and three corners of the hexagon are exactly the midpoints of the triangle's edges, so each hexagon touches its triangle at those three points.

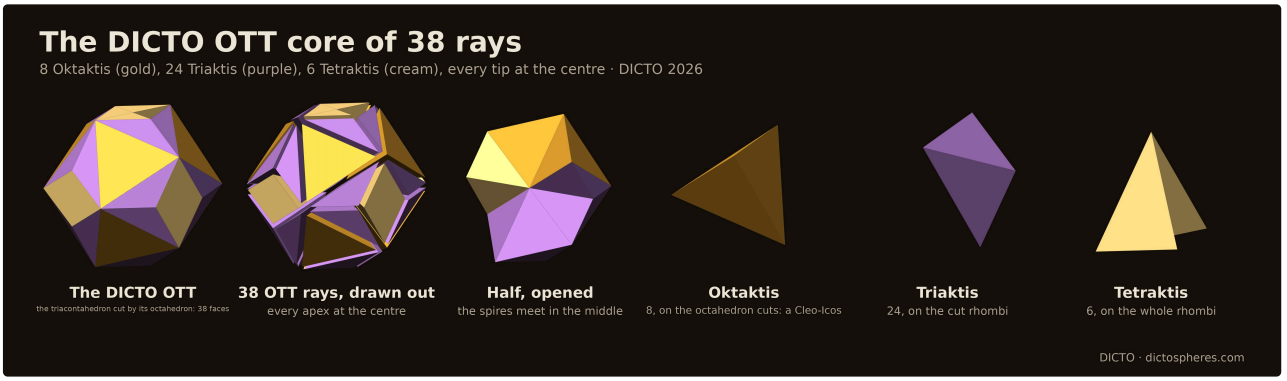


Figure 4. The OTT core: the DICTO OTT; its 38 rays drawn out, every tip at the centre; half of them, opened; one Oktaktis (gold), one Triaktis (purple) and one Tetraktis (cream).

6. The OG turned out: the golden step (exact)

The idea is DICTO's: turn the inside towers out and start a new sequence. Each of the 80 Aktis is flipped outward over its own face, and DICTO chose the **golden step**: the Aktis shortened to 1/φ of its height before it is flipped, so its tip lies at φ times the foot point of its face plane (the point of the plane nearest the centre). The next solid is the hull of the 80 tips; on it the same is done again, one beat at a time.

What it builds. The hull of the tips of a solid S is φ·P(S), where P is the reverse step of [2], and P commutes with scaling. Hence beat k is **φ^k times the k-th solid of the OG's downward family** [2]: the OG, φ·TPH, φ²·TQH, φ³·TQP, φ⁴·TQHD (DICTO's Triangle-Pentagon Hecatontatriacontadihedron, Triangle-Quadrilateral-Hexagon Polyhedron, Triangle-Quadrilateral-Pentagon Polyhedron and Triangle-Quadrilateral-Hexagon Diacosihedron). This is checked exactly: at each beat the corners are exactly the flipped tips, and each solid is fixed by the 60 rotations of the icosahedron. In the frame of [2] (ID circumradius φ):

Beat	Solid	Corners	Corner radii ≈
0	OG	102	1.618034, 1.687285
1	φ·TPH	80	2.445695, 2.564186
2	φ²·TQH	132	3.896825, 3.933014
3	φ³·TQP	140	6.003187, 6.032557, 6.324828
4	φ⁴·TQHD	162	9.248108, 9.555175, 9.560121

Nesting. Every solid lies inside the next (exact), so the sequence nests at every beat built, also from φ²·TQH to φ³·TQP; the downward family itself stops nesting at the TQH and TQP [2]. Beat k is exactly φ^k times the k-th downward solid, but the downward solids themselves shrink [2], so the outer radius grows by a little less than φ per beat: 1.52, 1.53, 1.61 and 1.51.

Towers. The shells of 1, 1 and 2 beats are cut into towers as in §1:

Shell	Beats	From → to	Towers	Each tower ≈	Shell volume ≈
1	1	OG → φ·TPH	60 on kites 20 on hexagons	0.349923 1.282073	46.6369
2	1	φ·TPH → φ²·TQH	60 on triangles 60 on triangles 12 on pentagons	1.003994 0.992512 3.727717	164.5229
3	2	φ²·TQH → φ⁴·TQHD	60 on triangles 60 on trapezoids 20 on hexagons	36.321214 13.483680 6.977223	3127.8381

Table 4. The OG turned out, golden step, in the frame of [2]; shell 3 passes through φ³·TQP. All values are exact in golden-exact.json (for instance a hexagon tower of shell 1 is (143097 + 71747√5)/236748); each shell's towers add up to the shell with remainder exactly 0.

So the shells hold **80, 132 and 140 towers** of **2, 3 and 3 shapes**. In shell 2 the 120 triangles of the TPH carry towers of two volumes; split by the 60 rotations they form two orbits of 60, each of one volume, so they are two shapes (60 + 60), and with the 12 pentagon towers the shell has three.

The full flip (float, for comparison). Flipping the Aktis at their whole height puts each tip at 2 times the foot point. The same argument gives beat k = 2^k times the k-th downward solid, and a float run agrees: 102, 80, 132, 140 and 162 corners at beats 0–4, the outer radius growing by about 1.9 per beat (flip-float.json). The golden step gives the same solids, growing by a little less than φ.

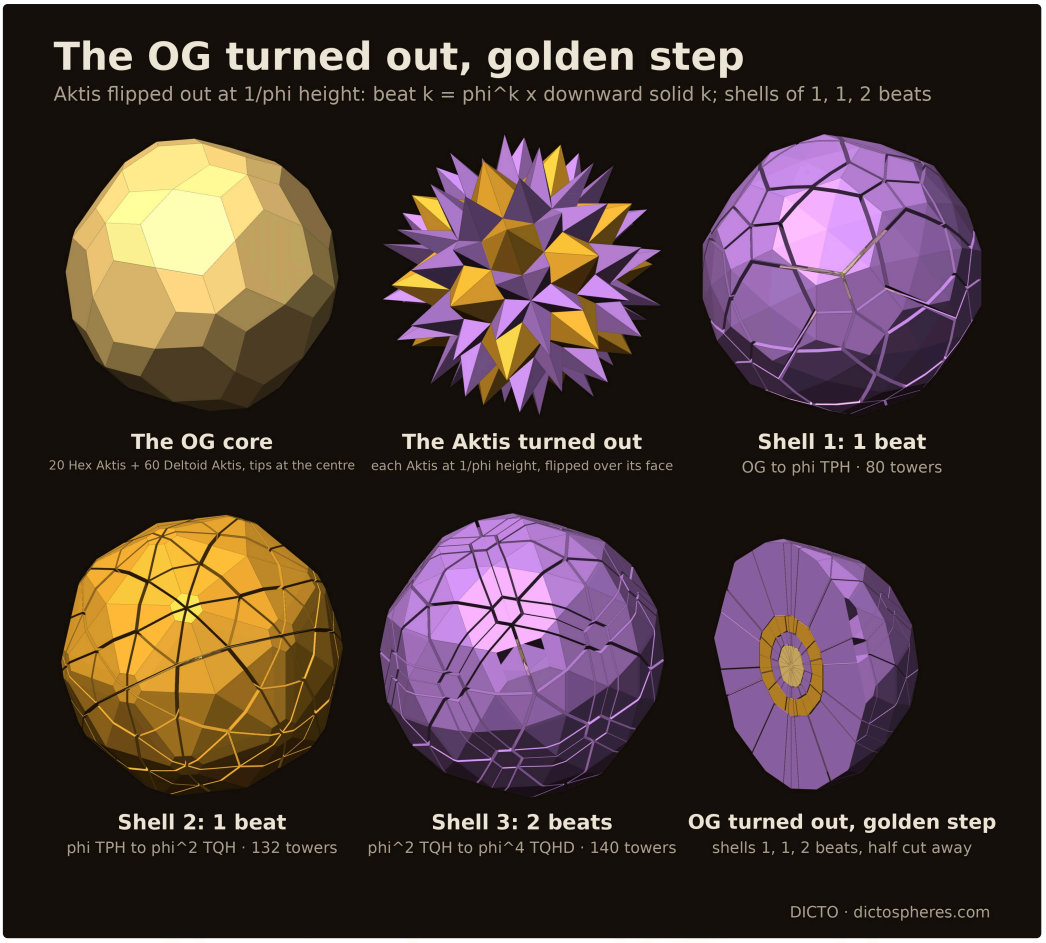


Figure 5. The OG turned out, golden step. The OG core (cream); the 80 Aktis at 1/φ of their height, flipped over their faces (Hex Aktis gold, Deltoid Aktis purple); shells 1, 2 and 3 (1, 1 and 2 beats) on their inner solids, purple, gold and purple; all three with half cut away. "phi TPH" is φ·TPH, and so on.

Shells 4 and 5 (pending). The shells of 3 and 5 beats, from beat 4 to beat 7 and from beat 7 to beat 12, need the downward solids of [2] beyond the fourth, built exactly; that computation is still running. Their solids, nesting, towers and volumes will be added in a later version; no numbers are given for them here.

Figure 6 (pending). Shells 4 and 5 of the OG turned out (3 and 5 beats, beats 4 → 7 → 12), to follow when the exact run is complete.

7. The surface patterns and the lines (exact)

DICTO observed that every shell so far has a different pattern on its surface, and that "the lines defining the shells denote the structures". We read each boundary solid two ways, both exactly: by its faces (the orbits of the 60 rotations, and how faces of one kind join edge to edge into groups) and by its lines (the edge network). The boundaries are few solids: the Cleo Towers end at ID | RT | GTI | ID | FPD | ID (each $10/\varphi^4$ to a power larger), the Polar Cleo Towers at RT | GTI | RT | FPD | ID | RT, and the OG turned out at OG | TPH | TQH | TQHD (each φ to a power larger; the TQP is passed through inside shell 3). Scale does not change a pattern, so Table 5 lists the nine solids once each.

Solid	Faces, and how faces of one kind group (exact)	Lines: edges; directions (exact)
ID	20 equilateral triangles, apart, on the three-fold axes; 12 regular pentagons, apart, on the five-fold axes	60; 15 directions of 4 edges, all along two-fold axes; 6 central belts of 10 edges
RT	30 rhombi, all joined	60; 6 directions of 10 edges, along the five-fold axes
GTI	12 regular pentagons, apart, on the five-fold axes; 20 hexagons, all joined	90; 15 directions of 6, along two-fold axes
FPD	60 isosceles triangles, all joined	90; 45 directions of 2 (15 along two-fold axes)
OG	60 kites in 12 groups of 5 on the five-fold axes; 20 regular hexagons, apart, on the three-fold axes, each bordered by kites on all six edges	180; 60 directions (30 of 2, 30 of 4), none along an axis
TPH	60 isosceles triangles, apart; 60 isosceles triangles in 30 pairs on the two-fold axes; 12 regular pentagons, apart, on the five-fold axes	210; 75 directions (15 of 6 along two-fold axes, 60 of 2)
TQH	60 isosceles triangles in 12 groups of 5 on the five-fold axes; 60 isosceles trapezoids in **30 ribbons** on the two-fold axes; 20 regular hexagons, apart, on the three-fold axes, each bordered by trapezoids on all six edges	270; 75 directions (15 of 6 along two-fold axes: the ribbons' lines; 30 of 2, 30 of 4)
TQP	60 isosceles triangles, apart; 30 rhombi, apart, on the two-fold axes; 60 isosceles trapezoids in 12 rings of 5 on the five-fold axes; 12 regular pentagons, apart, on the five-fold axes	300; 75 directions (15 of 8 along two-fold axes, 30 of 2, 30 of 4)
TQHD	60 isosceles triangles, apart; 60 isosceles triangles in 12 groups of 5 on the five-fold axes; 20 regular hexagons, apart, on the three-fold axes; 60 isosceles trapezoids in 12 rings of 5 on the five-fold axes	360; 105 directions (15 of 8 along two-fold axes, 60 of 2, 30 of 4)

Table 5. The boundary solids. "Apart": no two faces of that kind share an edge. A group is "on" an axis when a rotation about that axis maps it to itself. A direction is "along" an axis when its edges are parallel to it. A central belt is a closed circuit of edges in one plane through the centre.

Faces. No two boundaries of one sequence in a row have the same faces. In the OG's sequence each boundary solid has its own grouping: kites in fives (OG), triangle pairs on the two-fold axes (TPH), trapezoid ribbons on the two-fold axes (TQH), and trapezoid rings of five on the five-fold axes (TQHD), which repeat the rings of the TQP passed through inside shell 3.

Lines. No edge runs on straight through a corner, as convexity requires (a corner never lies between two points of the solid). The lines that line up across the surface are the parallel families. On every solid but the RT and the OG, 15 families point along the two-fold axes (on the RT, a zonohedron, the 6 families point along the five-fold axes); the OG alone has no edge along any axis. Only the ID has central belts: its six decagonal equators, as is classical.

The ribbons (DICTO's observation). On the TQH of the OG turned out (beat 2, $\varphi^2 \cdot$TQH) the 60 four-sided faces are congruent isosceles trapezoids, with parallel sides of squared length $(38367 + 17019\sqrt{5})/13448$ and $(6615 + 2925\sqrt{5})/3362$ (≈ 2.3839 and 1.9781), legs ≈ 0.4343, corner angles $\approx 62.15°$ and $117.85°$ and height ≈ 0.3840. They pair up across their shorter parallel sides: 30 shared edges, every trapezoid in exactly one pair, each pair centred on a two-fold axis (a half-turn about it swaps the two), two pairs per axis, one at each end. So each ribbon is drawn by three parallel lines, its two outer sides and the seam, bent $\approx 23.28°$ across the seam (float). The 15 edge families of the TQH that point along a two-fold axis are exactly these lines: 6 per family, the two ribbons at the two ends of one two-fold axis, all running along a second two-fold axis. Shell 3 stands a long blade-like tower on each trapezoid, so each ribbon carries two blades (Figure 7). In the TQP and the TQHD the trapezoids join along their legs instead, into rings of five on the five-fold axes (each trapezoid meeting two others); of the nine solids only the TQH has ribbons. DICTO sees in the ribbons a possible new shell structure; we record the observation and claim nothing beyond the checks above.

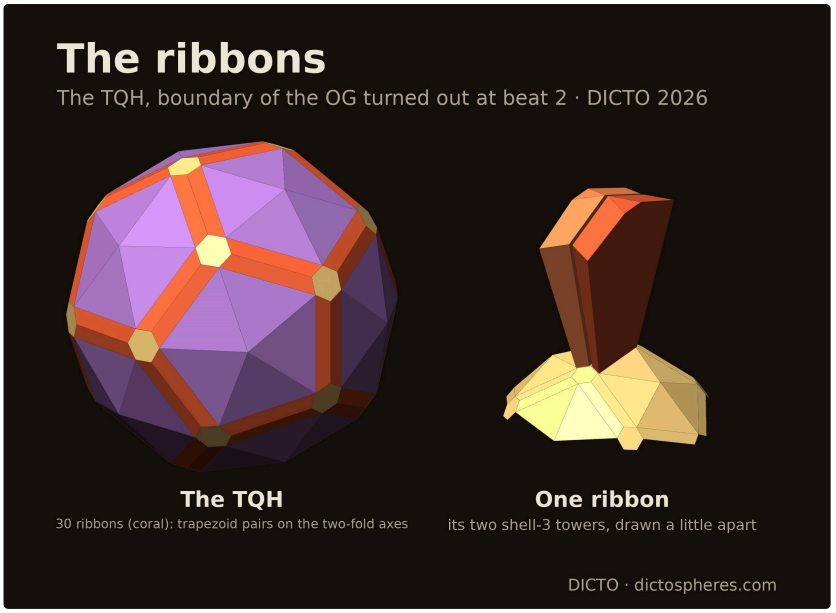


Figure 7. The ribbons. Left: the TQH (beat 2 of the OG turned out), triangles purple, hexagons cream, the 30 ribbons of trapezoid pairs coral. Right: one ribbon with its two shell-3 towers (coral, drawn a little apart across the seam), on the TQH's nearby faces (cream).

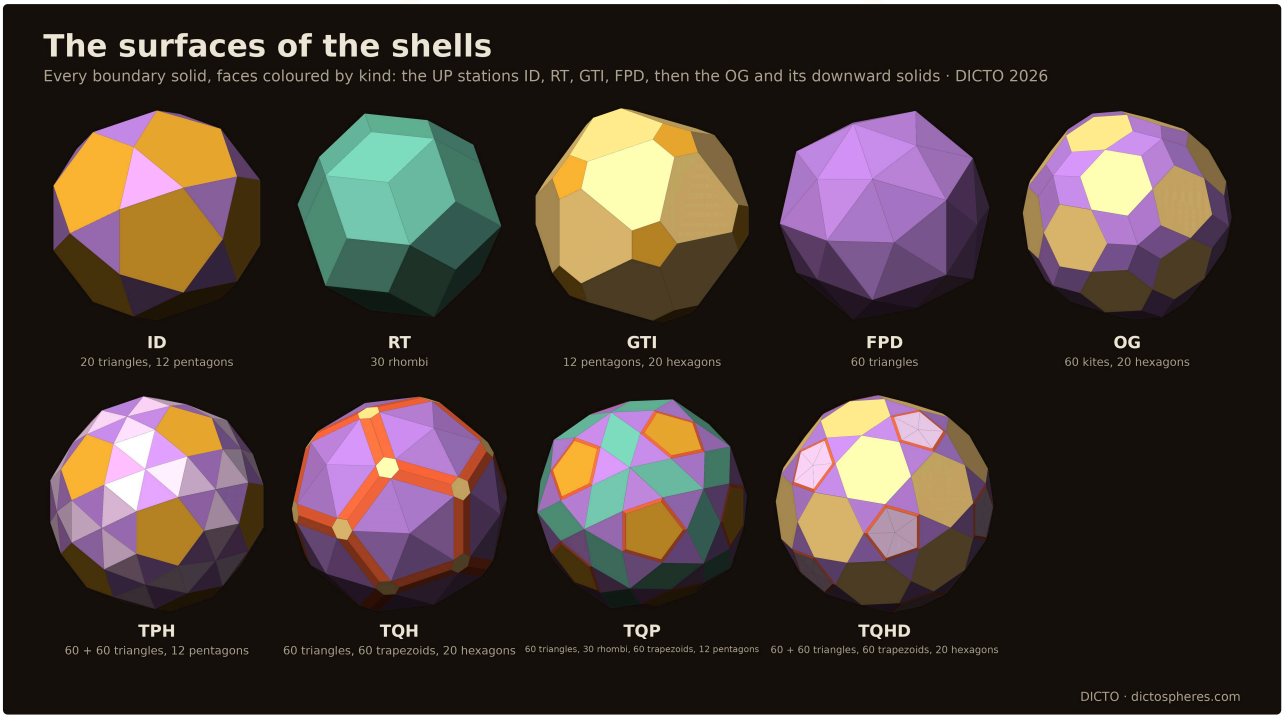


Figure 8. The surfaces of the shells: the nine boundary solids, all seen the same way, each to its own size. Faces coloured by kind: triangles purple (a second kind of triangle lilac), pentagons gold, hexagons cream, kites purple, rhombi teal, trapezoids coral.

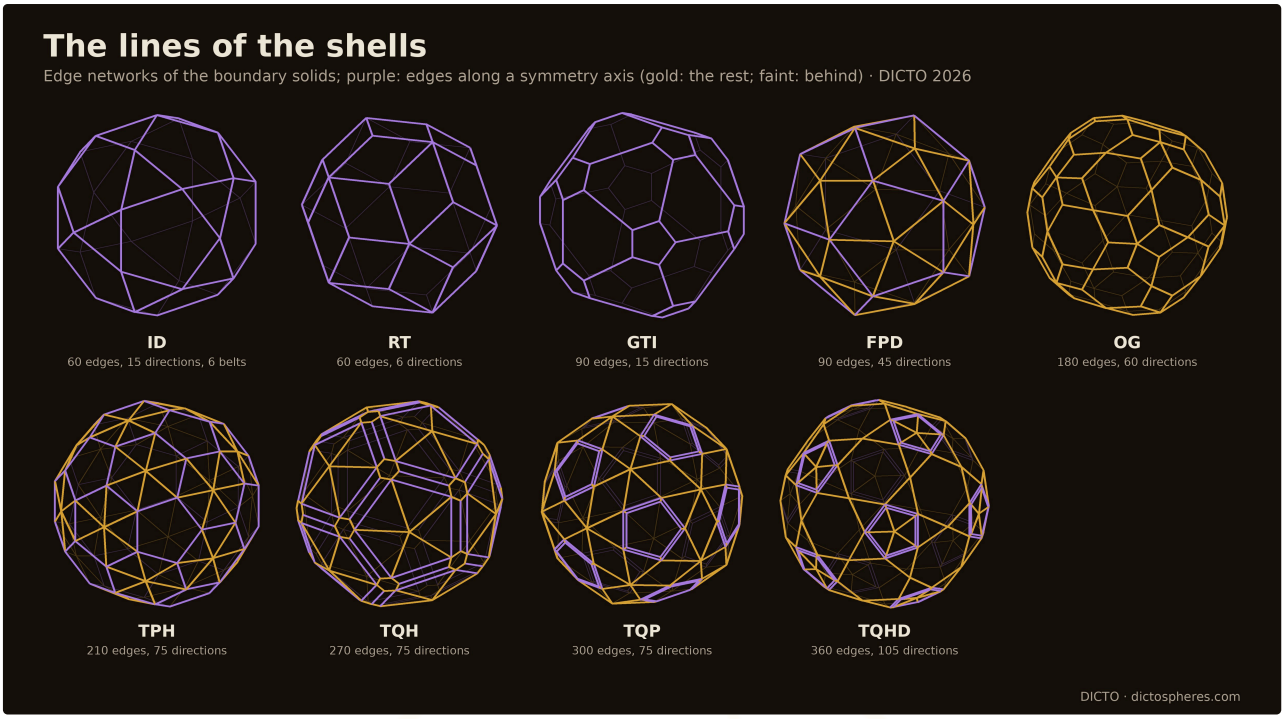


Figure 9. The lines of the shells: the edge network of each boundary solid, seen as in Figure 8. Purple: the edges that point along a symmetry axis (on the ID, the RT and the GTI all of them); gold: the rest; faint: the lines behind.

8. Prior art and what is claimed

- **Known and credited.** The step T is the three-dimensional antipedal construction about the centre, and its reverse P is Smarandache's face pedal polyhedron [9, 10] (as recorded in [1, 2]); reciprocation about a sphere is classical [17]. Solids nested round one centre go back to Kepler's *Mysterium cosmographicum* [11]; icosahedral shells round a centre are found in Mackay's packing of equal spheres [12], in the clusters of icosahedral quasicrystals [13] and in the nested icosahedral point sets of viruses and carbon onions [14, 15]; Bergman's continuous spherical truss surrounds a core of twenty tetrahedral pyramids with endless layers [16]; Kramer filled space round a centre with cells that recur τ^3 larger [18]. Cutting a solid into pyramids from its centre is as old as Hypsicles' "Book XIV" of Euclid, whose icosahedron pyramids are the Cleo-Icos [7, 19]; the Cleo-Icos is also the unit tetrahedron of Mackay's icosahedron [7, 12]. The Fibonacci numbers and the golden ratio are classical.
- **Already published by DICTO.** The UP rule, its four stations and its first shells' pieces, and the chain's self-polarity [1]; the OG [3] and its downward family [2]; the OTT [5]; CLEO [6]; the CLEO sphere and the Cleo-Icos [7] (all listed in [8]). Shells 1 and 2 of the Cleo Towers and shell 4 of the Polar Cleo Towers are shells of [1].
- **Claimed here:** the Fibonacci fusing of beats into tower shells and the exact towers of the fused shells (Cleo Towers, shells 3–5); the Polar Cleo Towers as exact tower shells, with the Fibonacci boundaries carried by the polarity of [1]; the OG core of 80 Aktis and the OTT core of 38 OTT rays as exact sets of one shape per kind, with their shares; the Oktaktis as Cleo-Icos lying in the OG's hexagon planes; and the OG turned out with the golden step: beat $k = \varphi^k$ times the k-th downward solid, the nesting at every beat built, and the exact towers of its first three shells; the surface patterns and edge networks of the boundary solids (Table 5), including the TQH's 30 ribbons.

In the sources checked (the full text of the 77 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's link-only entries were not searched) we did not find tower shells in Fibonacci beats, the Polar Cleo Towers, the Aktis or ray cores of the OG and the OTT, the OG turned out, or the ribbons of the TQH.

9. Further work

- Shells 4 and 5 of the OG turned out (3 and 5 beats), exact (§6), and whether the nesting continues.

- The full flip (tip = 2 × foot point) built exactly, and its shells.
- The surface patterns and lines of the shells 4 and 5 boundaries of the OG turned out, when built.

Outlook (DICTO's idea, untested). DICTO notes that these shells may mirror material structures, as geodesic forms did before C_{60} [24] and carbon nanotubes [25] were found. Nested icosahedral shells do occur in nature: in the clusters of quasicrystals and their approximants (Mackay [12], Bergman [22], Tsai [13]), in metal and boron nanoclusters, and in virus capsids (Caspar and Klug [23]; see also [26]). Whether the shells of this paper correspond to any real structure has not been tested; a checkable next step is to compare their corner counts and radius ratios with published sequences of cluster shells.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$), with the tools of [1] (exact_cycle.py, exact_shells.py, exact_downward.py in studies/2026-10-19-five-fold-octet) and of [4] (q5.py, exactpoly.py in studies/2026-10-18-ocd-octet/scripts): tower_shells_exact.py (the Cleo Towers; log tower-shells-exact.log, 6 PASS; data tower-shells-exact.json) and polar_exact.py (the polarity and the Polar Cleo Towers; polar-exact.log, 8 PASS; polar-exact.json) in 2026-10-20-cleo-towers; og_spires_exact.py (the 80 Aktis; og-spires-exact.log, 7 PASS) in 2026-10-20-og-spires; ott_rays_exact.py (the 38 rays; ott-rays-exact.log, 5 PASS) and ott_og_planes.py (the octahedral cuts in the OG's hexagon planes; ott-og-planes.log, 4 PASS, also the hexagon touching its triangle), on the OTT of [5] (cut8-exact.json, cut8-exact.log), in 2026-10-20-ott-rays; golden_exact.py (the OG turned out; golden-exact.log, 15 PASS and one FAIL line, which is only the test "one shape per kind of face" for shell 2, where the triangle towers have two shapes) and golden_orbits.py (shell 2 by symmetry orbit: two triangle orbits of 60 and one pentagon orbit of 12, one volume each; golden-orbits.log, 6 PASS) golden_ribbons.py (the trapezoids of the TQH, TQP and TQHD and how they join; golden-ribbons.log, 14 PASS), surface_patterns.py (faces, kinds and groups of the nine boundary solids; surface-patterns.log, 28 PASS) surface_lines.py (their edge networks; surface-lines.log, 12 PASS) and kyushindo_checks.py (the condition $w \cdot v \le w \cdot w$ for every station, and the 60-rotation symmetry of the inner and outer solid of every shell; kyushindo-checks.log, 13 PASS), in 2026-10-21-og-turned-out. Float: flip_float.py (flip-float.json). Each script takes the path of its tool folder as its argument (python3 <script> <tool folder>); polar_exact.py chooses the corner triples by an exact determinant where the published helper of [1] used a float test. Figures: *_figs.py, shots.mjs and *_compose.py in each folder (Figures 7–9: surface_figs.py and surface_compose.py, which also draws the line figure) (float copies of the exact solids); the cover: cover.py in this paper's folder.

The series name

Kyushindo is the martial art of Kenshiro Abbe [20, 21]. DICTO gives its three principles as Change 万物流転, Rhythm 律動 and Harmony 調和, and his brief for this series was "rhythm, harmony and change in the construction of towers and cores". Here rhythm is the Fibonacci beats, harmony the exact golden fits, and change that each shell's form differs from the last (§7).

References

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2. DICTO, The DICTO OG Family: Four Steps Up and Four Down, DICTO DOI: 10.5281/zenodo.23488944.
3. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
4. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the frame, and the exact tools q5.py and exactpoly.py).
5. DICTO, The DICTO Ogdoecontahedron on the Gyrated Octet, DICTO DOI: 10.5281/zenodo.23484012 (the DICTO OTT).
6. DICTO, AXE, CLEO and FUJI, DICTO DOI: 10.5281/zenodo.23351291.
7. DICTO, The FUJI Family, DICTO DOI: 10.5281/zenodo.23357896 (§5.5 the CLEO sphere, §5.6 the Cleo-Icos).
8. DICTO, Collected Geometric Works, DICTO DOI: 10.5281/zenodo.23345699.
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21. Kenshiro Abbe, Wikipedia, https://en.wikipedia.org/wiki/Kenshiro_Abbe (saved 11 October 2026).
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Appendix. The reference library searched

The library's manifests list 261 entries, of which 90 have a local file; with three files at its root, these give the 77 distinct sources with a text layer listed below and 13 page images. The full text of the 77 sources was searched on 21 October 2026 for: pedal, antipedal, reciprocation, polar, nested, concentric, shell, Fibonacci, central pyramid, spire, tower, ogdoecontahedron, octahedrally truncated, Hypsicles, Mackay and Kyushindo. None contains the results of this paper. The relevant hits are Kepler's nested solids [11]; the nested icosahedral point sets of Dechant, Wardman, Keef and Twarock and of Twarock, Valiunas and Zappa [14, 15]; the concentric cluster shells of Fujita, Takano, Yamamoto and Tsai [13]; Fibonacci sequences in Baake's guide and the other aperiodic-order surveys; and the Wikipedia pages on Kyūshindō and Kenshiro Abbe [20, 21]. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the about 170 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a

		Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times

M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)